

Linear Detection Based on Block-Wise Low-Rank Matrix Approximation for XL-MIMO Systems

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Abstract—The high dimensionality of antenna arrays in extremely large-scale multiple-input-multiple-output (XL-MIMO) systems leads to prohibitive computational complexity in uplink linear detection. To exploit the intrinsic low-rank structure of near-field channels and enable low-complexity linear detection, the global low-rank matrix fitting (LMaFit) approach with a rank-increasing strategy is employed to facilitate the matrix computation, where the rank adaptation leverages the underlying propagation structure. Building upon it, a block-wise LMaFit scheme is further proposed to utilize localized low-rank structures within subarrays, thereby enabling substantial complexity reduction in linear detection. Finally, the convergence behaviors of both the global and block-wise LMaFit algorithms are analyzed.

Index Terms—XL-MIMO, signal detection, low-rank matrix approximation, near-field, visibility region.

I. INTRODUCTION

Extremely large-scale multiple-input-multiple-output (XL-MIMO) has emerged as a promising technology for the future wireless systems [1]. With increasing antenna array aperture, XL-MIMO pushes the electromagnetic propagation from far-field region with planar wave assumptions to the near-field regime characterized by spherical wave propagation. This results in spatial non-stationarity and visibility region (VR) effects, where each user is only visible to a subset of antennas [2]. Additionally, the dramatically increased antenna array size in XL-MIMO systems places a pressing challenge on uplink signal detection in terms of computational complexity.

Linear detection such as zero-forcing (ZF) and minimum mean-square error (MMSE) detectors are widely adopted due to their favorable performance. However, their practical implementation is demanding owing to the complicated matrix multiplication. More importantly, conventional linear detectors treat the channel matrix as full dimension and fail to exploit the intrinsic low-rank structure arising from limited propagation paths, leading to substantial computational redundancy. To exploit the low-rank structure of the channel, truncated singular value decomposition (TSVD) is a natural approach but incurs prohibitive SVD complexity and requires prior knowledge of the truncation rank. As a low-complexity alternative, low-rank matrix fitting (LMaFit) provides an efficient low-rank matrix approximation without resorting to SVD [3]. In [4], by using the spatial correlation of the full-dimensional massive MIMO channels, LMaFit has been

applied to reduce the complexity of the Gram matrix computation under far-field scenarios. Nevertheless, extending such low-rank techniques to XL-MIMO systems under spatial non-stationarity remains a challenging problem.

In this paper, we develop an LMaFit-based linear detection for near-field XL-MIMO systems. First, a global LMaFit scheme is developed to enable low-complexity Gram matrix construction in linear detection, where a rank-increasing strategy is incorporated to adaptively exploit the intrinsic low-rank structure of the channel without requiring prior rank knowledge. Furthermore, to better adapt the global framework to spatially non-stationary near-field channels, a block-wise LMaFit strategy is proposed that partitions the antenna array into multiple subarrays, leading to low-rank approximation problems with smaller dimensions and reduced approximation ranks. Finally, the computational complexity and convergence behaviors are analyzed, and simulation results confirm the improved trade-off between performance and complexity of the proposed LMaFit-based linear detection.

II. SYSTEM MODEL

We consider an XL-MIMO uplink system where the base station (BS) is equipped with a uniform planar array (UPA) of $N = M_x M_z$ antennas serving K single-antenna users. Here, M_x and M_z are the numbers of antennas along the horizontal and vertical dimensions, respectively, with antenna spacings dx and dz . The channel of the k -th user is denoted by the matrix $\mathbf{H}_k = [\mathbf{h}_{k,1}, \mathbf{h}_{k,2}, \dots, \mathbf{h}_{k,i}, \dots, \mathbf{h}_{k,M_x}] \in \mathbb{C}^{M_z \times M_x}$, where $\mathbf{h}_{k,i} \in \mathbb{C}^{M_z \times 1}$. The vectorized form of $\mathbf{H}_k$ is

$$\mathbf{h}_k^{\text{vec}} = \text{vec}\{\mathbf{H}_k\}. \quad (1)$$

We adopt the multipath near-field channel model based on the spherical wave assumption [5], under which the channel matrix $\mathbf{H}_k$ can be expressed as

$$\mathbf{H}_k = \left(\sum_{l=1}^L \gamma_{k,l} \mathbf{a}_x^T(\theta_{k,l}, \phi_{k,l}, r_{k,l}) \otimes \mathbf{a}_z(\phi_{k,l}, r_{k,l}) \right) \odot \mathbf{D}_k. \quad (2)$$

Here, L is the number of propagation paths, and $\gamma_{k,l}$ is the complex gain of the l -th path associated with the k -th user. The path gains $\{\gamma_{k,l}\}_{l=1}^L$ are independent and identically distributed (i.i.d.) complex Gaussian random variables with zero-mean and variance $1/L$. Consequently, $\mathbf{H}_k$ exhibits the low-rank structure when $L < \min(M_z, M_x)$, due to the limited number of

propagation paths. $\mathbf{D}_k \in \{0, 1\}^{M_z \times M_x}$ represents the VR mask, and is assumed to be known for algorithm design. The vectors $\mathbf{a}_z(\phi_{k,l}, r_{k,l}) \in \mathbb{C}^{M_z}$ and $\mathbf{a}_x(\theta_{k,l}, \phi_{k,l}, r_{k,l}) \in \mathbb{C}^{M_x}$ denote the near-field array response vectors along the vertical and horizontal dimensions, respectively. The n -th entry of $\mathbf{a}_x(\theta_{k,l}, \phi_{k,l}, r_{k,l})$ and m -th entry of $\mathbf{a}_z(\phi_{k,l}, r_{k,l})$ are given by

$$\begin{aligned} [\mathbf{a}_x(\theta_{k,l}, \phi_{k,l}, r_{k,l})]_n &= e^{-j \frac{2\pi}{\lambda} \left(-nd_x \cos \theta_{k,l} \sin \phi_{k,l} + \frac{n^2 d_x^2 (1 - \cos^2 \theta_{k,l} \sin^2 \phi_{k,l})}{2r_{k,l}} \right)}, \\ [\mathbf{a}_z(\phi_{k,l}, r_{k,l})]_m &= e^{-j \frac{2\pi}{\lambda} \left(-md_z \cos \phi_{k,l} + \frac{m^2 d_z^2 \sin^2 \phi_{k,l}}{2r_{k,l}} \right)}, \end{aligned} \quad (3)$$

where $n = \tilde{n} - \frac{M_x}{2}$ with $\tilde{n} \in \{0, \dots, M_x - 1\}$ and $m = \tilde{m} - \frac{M_z}{2}$ with $\tilde{m} \in \{0, \dots, M_z - 1\}$, and $r_{k,l}$ is the distance between the user antenna and the BS arrays center. To account for spatial non-stationarity, the $M_z \times M_x$ UPA is uniformly partitioned into $N_z \times N_x$ subarrays, each with $G_z \times G_x$ antennas, where $G_z = M_z/N_z$ and $G_x = M_x/N_x$. Each user is assumed to be visible to B subarrays, where typically $B \ll N_z N_x$ (e.g., $B/N_z N_x = 1/8$). The VR mask $\mathbf{D}_k \in \{0, 1\}^{M_z \times M_x}$ is defined as

$$[\mathbf{D}_k]_{m_z, m_x} = \begin{cases} 1, & \left(\left\lceil \frac{m_z N_z}{M_z} \right\rceil, \left\lceil \frac{m_x N_x}{M_x} \right\rceil \right) \in \tilde{\mathcal{S}}_k, \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $\tilde{\mathcal{S}}_k$ is the index set of B visible subarrays with $|\tilde{\mathcal{S}}_k| = B$. Consequently, $\mathbf{D}_k$ contains $B G_z G_x$ non-zero entries. Therefore, the received signal vector $\mathbf{y} \in \mathbb{C}^N$ at BS can be represented as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (5)$$

where $\mathbf{x} \in \mathcal{X}^K$ is the transmitted vector from the discrete complex M -QAM constellation $\mathcal{X}^K$, and $\mathbf{n} \in \mathbb{C}^N$ is the additive white Gaussian noise (AWGN) with covariance matrix $\sigma_n^2 \mathbf{I}_N$ and zero mean. The overall channel matrix is formed by stacking the vectorized channel matrices of all users as

$$\mathbf{H} = [\mathbf{h}_1^{\text{vec}}, \dots, \mathbf{h}_K^{\text{vec}}] \in \mathbb{C}^{N \times K}. \quad (6)$$

Given the system model in (5), to recover the transmitted signal $\mathbf{x}$, the traditional linear detection schemes like ZF and MMSE are commonly applied due to the significant detection performance, which work as follows

$$\tilde{\mathbf{x}}_{\text{ZF}} = ((\mathbf{H})^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}, \quad (7)$$

$$\tilde{\mathbf{x}}_{\text{MMSE}} = ((\mathbf{H})^H \mathbf{H} + \sigma_n^2 \mathbf{I}_K)^{-1} (\mathbf{H})^H \mathbf{y}, \quad (8)$$

respectively. Then, the signal detection $\hat{\mathbf{x}}$ is determined by quantizing $\hat{\mathbf{x}}_{\text{ZF}} = \lceil \tilde{\mathbf{x}}_{\text{ZF}} \rceil_{\mathcal{Q}} \in \mathcal{X}^K$ or $\hat{\mathbf{x}}_{\text{MMSE}} = \lceil \tilde{\mathbf{x}}_{\text{MMSE}} \rceil_{\mathcal{Q}} \in \mathcal{X}^K$. However, ZF and MMSE detectors become prohibitive in XL-MIMO because of the intensive matrix operations involved. Specifically, the computational complexity primarily stems from two parts: the matrix multiplication $(\mathbf{H})^H \mathbf{H}$ with computational complexity $\mathcal{O}(NK^2)$, and the inversion of the resulting matrix with computational complexity $\mathcal{O}(K^3)$ [6]. In typical XL-MIMO scenarios where $N \gg K$, the $\mathcal{O}(NK^2)$ term dominates, thereby limiting practical implementation.

III. LOW-COMPLEXITY GRAM MATRIX FORMULATION BASED ON LOW-RANK MATRIX APPROXIMATION

In this section, we present a low-complexity Gram matrix formulation that exploits the spatial low-rank structure of the channel matrix. To achieve it, the global LMaFit with an increasing rank strategy is employed to obtain the required low-rank representation, followed by a comprehensive analysis of the overall computational complexity.

A. Global LMaFit-Based Low-Rank Channel Approximation

According to (2), the subchannel matrix $\mathbf{H}_k$ exhibits an inherently low-rank structure along both horizontal and vertical dimensions due to spatial non-stationarity. Motivated by this property, we approximate the subchannel matrix $\mathbf{H}_k$ by a separable rank- $\tilde{r}$ representation given by

$$\mathbf{H}_k \approx \mathbf{H}_{k,z} \mathbf{H}_{k,x}^H, \quad (9)$$

where

$$\begin{aligned} \mathbf{H}_{k,z} &= [\mathbf{h}_{k,z,1}, \dots, \mathbf{h}_{k,z,r}] \in \mathbb{C}^{M_z \times \tilde{r}}, \\ \mathbf{H}_{k,x} &= [\mathbf{h}_{k,x,1}, \dots, \mathbf{h}_{k,x,r}] \in \mathbb{C}^{M_x \times \tilde{r}} \end{aligned} \quad (10)$$

contain $\tilde{r}$ dominant vertical and horizontal spatial modes of $\mathbf{H}_k$, respectively. Based on the separable low-rank model in (9), the complexity of matrix multiplication in XL-MIMO systems can be significantly reduced.

Accordingly, we employ the LMaFit algorithm to compute a rank- $\tilde{r}$ approximation $\hat{\mathbf{H}}_k$ of $\mathbf{H}_k$ by solving the following factorized optimization formulation [3]

$$\min_{\hat{\mathbf{H}}_{k,z}, \hat{\mathbf{H}}_{k,x}} F(\hat{\mathbf{H}}_{k,z}, \hat{\mathbf{H}}_{k,x}), \quad (11)$$

where

$$F(\hat{\mathbf{H}}_{k,z}, \hat{\mathbf{H}}_{k,x}) = \frac{1}{2} \left\| \mathbf{H}_k - \hat{\mathbf{H}}_{k,z} \hat{\mathbf{H}}_{k,x}^H \right\|_F^2. \quad (12)$$

Unlike TSVD, LMaFit avoids computing full or partial SVDs and directly optimizes the factor matrices $\hat{\mathbf{H}}_{k,z}$ and $\hat{\mathbf{H}}_{k,x}$ through the following alternating minimization procedure. Specifically, for the given rank $\tilde{r}$, the factor matrix $\hat{\mathbf{H}}_{k,x}$ is initialized as

$$\hat{\mathbf{H}}_{k,x}^{(0)} = \begin{bmatrix} \mathbf{I}_{\tilde{r}} \\ \mathbf{0}_{(M_x - \tilde{r}) \times \tilde{r}} \end{bmatrix} \in \mathbb{C}^{M_x \times \tilde{r}}. \quad (13)$$

At iteration $t+1$, given $\hat{\mathbf{H}}_{k,x}^{(t)}$, the update of $\hat{\mathbf{H}}_{k,z}$ is obtained by

$$\hat{\mathbf{H}}_{k,z}^{(t+1)} = \mathbf{H}_k \hat{\mathbf{H}}_{k,x}^{(t)} \left(\hat{\mathbf{H}}_{k,x}^{(t)H} \hat{\mathbf{H}}_{k,x}^{(t)} \right)^{-1}. \quad (14)$$

Subsequently, with $\hat{\mathbf{H}}_{k,z}^{(t+1)}$ fixed, $\hat{\mathbf{H}}_{k,x}$ is updated as

$$\hat{\mathbf{H}}_{k,x}^{(t+1)} = \mathbf{H}_k^H \hat{\mathbf{H}}_{k,z}^{(t+1)} \left(\hat{\mathbf{H}}_{k,z}^{(t+1)H} \hat{\mathbf{H}}_{k,z}^{(t+1)} \right)^{-1}. \quad (15)$$

The approximation quality at iteration t is characterized by the residual matrix

$$\mathbf{S}^{(t)} = \mathbf{H}_k - \hat{\mathbf{H}}_k^{(t)} = \mathbf{H}_k - \hat{\mathbf{H}}_{k,z}^{(t)} \hat{\mathbf{H}}_{k,x}^{(t)H}, \quad (16)$$

which reflects the spatial components that are not captured by the current low-rank matrix approximation $\hat{\mathbf{H}}_k^{(t)}$. We then adopt a rank adaptation strategy to adaptively update the rank based on the residual structure.

The rank is initialized as $\tilde{r}^{(0)} = 1$. At iteration t , a random vector $\mathbf{u}_0$ is generated and normalized. The dominant left singular vector $\mathbf{u}_S$ of $\mathbf{S}^{(t)}$ is approximated by one step of power iteration on $\mathbf{S}^{(t)}\mathbf{S}^{(t)H}$ as

$$\mathbf{v} = \mathbf{S}^{(t)H}\mathbf{u}_0, \quad \mathbf{u}_S = \frac{\mathbf{S}^{(t)}\mathbf{v}}{\|\mathbf{S}^{(t)}\mathbf{v}\|_2}. \quad (17)$$

Accordingly, the candidate covered region $\mathcal{V}_{\text{new}}$ is identified as the index set corresponding to the nonzero entries of $\mathbf{u}_S$:

$$\mathcal{V}_{\text{new}} = \{m : |\mathbf{u}_S(m)|^2 > 0\}. \quad (18)$$

Meanwhile, the region already captured by $\hat{\mathbf{H}}_k^{(t)}$ is characterized through a row-wise energy coverage criterion as

$$\mathcal{V}_{\text{old}} = \left\{ m : \frac{\sum_n |\hat{\mathbf{H}}_k^{(t)}(m, n)|^2}{\sum_n |\mathbf{H}_k(m, n)|^2} > \tau \right\}, \quad \tau = 0.9. \quad (19)$$

The rank evolution is then governed by whether the dominant residual component reveals previously uncovered spatial support:

$$\tilde{r}^{(t+1)} = \begin{cases} \tilde{r}^{(t)} + 1, & \text{if } \mathcal{V}_{\text{new}} \setminus \mathcal{V}_{\text{old}} \neq \emptyset, \\ \tilde{r}^{(t)}, & \text{otherwise.} \end{cases} \quad (20)$$

When the rank $\tilde{r}$ is increased, i.e., $\tilde{r}^{(t+1)} = \tilde{r}^{(t)} + 1$, the factor matrices are expanded by appending $\mathbf{u}_S$ as

$$\hat{\mathbf{H}}_{k,z}^{(t+1)} = [\hat{\mathbf{H}}_{k,z}^{(t)}, \mathbf{u}_S], \quad \hat{\mathbf{H}}_{k,x}^{(t+1)} = [\hat{\mathbf{H}}_{k,x}^{(t)}, (\mathbf{H}_k^H \mathbf{u}_S)^H]. \quad (21)$$

This expansion follows the dominant residual direction $\mathbf{u}_S$, which captures the most significant unexplained component, in contrast to random directions. To summarize, the proposed global LMaFit with rank adaptation scheme is outlined in Algorithm 1, where the QR-based updates are equivalent to the alternating least-squares steps in (14) and (15). Moreover, the convergence of the global LMaFit algorithm to a stationary point is proved in [3], as illustrated in the following Lemma.

Lemma 1 [3]. *The factor matrices $\{(\hat{\mathbf{H}}_{k,z}^{(t)}, \hat{\mathbf{H}}_{k,x}^{(t)})\}$ generated by LMaFit converge to a stationary point of the global optimization problem in (11) in the limit, i.e.,*

$$\lim_{t \rightarrow \infty} \nabla_{\hat{\mathbf{H}}_{k,z}^{(t)}, \hat{\mathbf{H}}_{k,x}^{(t)}} F(\hat{\mathbf{H}}_{k,z}^{(t)}, \hat{\mathbf{H}}_{k,x}^{(t)}) = \mathbf{0}. \quad (22)$$

B. Complexity Analysis

Here, the computational complexity is evaluated in terms of the required number of complex multiplications. For simplicity, we assume the same approximation rank $\tilde{r}$ for all K users in the complexity analysis.

Specifically, the overall computational complexity of the global LMaFit-based linear detection consists of the following three stages. During the first stage, LMaFit processing, the computational complexity of steps 3, 4 and 6 in Algorithm 1 are at the

Algorithm 1 Global LMaFit Algorithm

Input: $\mathbf{H}_k \in \mathbb{C}^{M_z \times M_x}$, T_{max}

Output: Approximated matrix $\mathbf{H}_k \approx \hat{\mathbf{H}}_{k,z} \hat{\mathbf{H}}_{k,x}^H$

- 1: **Initial:** $t = 0$, $\tilde{r}^{(0)} = 1$, $\hat{\mathbf{H}}_{k,x}^{(0)} = [1, 0, \dots, 0]^T \in \mathbb{C}^{M_x \times 1}$
- 2: **repeat**
- 3: form $\mathbf{Y} = \mathbf{H}_k \hat{\mathbf{H}}_{k,x}^{(t)} \in \mathbb{C}^{M_z \times \tilde{r}^{(t)}}$
- 4: QR decomposition: $\mathbf{Y} = \mathbf{U}\mathbf{R}$
- 5: update $\hat{\mathbf{H}}_{k,z}^{(t)} = \mathbf{U} \in \mathbb{C}^{M_z \times \tilde{r}^{(t)}}$
- 6: update $(\hat{\mathbf{H}}_{k,x}^{(t)})^H = (\hat{\mathbf{H}}_{k,z}^{(t)})^H \mathbf{H}_k \in \mathbb{C}^{\tilde{r}^{(t)} \times M_x}$
- 7: compute residual $\mathbf{S}^{(t)}$ according to (16)
- 8: identify $\mathcal{V}_{\text{new}}$ and $\mathcal{V}_{\text{old}}$ according to (18) and (19)
- 9: update the rank $\tilde{r}^{(t+1)}$ according to (20)
- 10: **if** $\tilde{r}^{(t+1)} > \tilde{r}^{(t)}$ **then**
- 11: update $\hat{\mathbf{H}}_{k,z}^{(t+1)}$ and $\hat{\mathbf{H}}_{k,x}^{(t+1)}$ according to (21)
- 12: **end if**
- 13: $t = t + 1$
- 14: **until** convergence or $t = T_{\text{max}}$

order of $\mathcal{O}(M_z M_x \tilde{r})$, $\mathcal{O}(M_z \tilde{r}^2)$ and $\mathcal{O}(M_z M_x \tilde{r})$, respectively. Subsequently, in Steps 7 and 8 of the Algorithm 1, we use a single power iteration to approximately compute the dominant left singular vector $\mathbf{u}_S$, resulting in the complexity of $\mathcal{O}(M_z M_x)$. Then calculating $\mathcal{V}_{\text{new}}$ and $\mathcal{V}_{\text{old}}$ requires $\mathcal{O}(M_z \tilde{r})$ operations. After that, the expansions of matrices $\hat{\mathbf{H}}_{k,z}$ and $\hat{\mathbf{H}}_{k,x}$ need complexities of $\mathcal{O}(M_z M_x)$.

In the second stage, Gram matrix construction is performed with substantially reduced computational complexity that becomes dependent on the rank $\tilde{r}$, as detailed in Proposition 1.

Propositon 1. *Based on the rank- $\tilde{r}$ factorized representation in (9) with $\tilde{r} < \min(M_z, M_x)$ for all K users, the computational complexity of the Gram matrix $\hat{\mathbf{G}} = \hat{\mathbf{H}}^H \hat{\mathbf{H}}$ can be reduced from $\mathcal{O}(M_z M_x K^2)$ to $\mathcal{O}(\tilde{r}^2 M_z K^2 + \tilde{r}^2 M_x K^2)$.*

Proof: The (i, j) -th elements of $\hat{\mathbf{G}}$ can be written by

$$\begin{aligned} \hat{\mathbf{G}}_{ij} &= \left(\text{vec}(\hat{\mathbf{H}}_i) \right)^H \text{vec}(\hat{\mathbf{H}}_j) \\ &= \sum_{l=1}^{\tilde{r}} \sum_{l'=1}^{\tilde{r}} \left(\text{vec}(\hat{\mathbf{h}}_{i,z,l} \hat{\mathbf{h}}_{i,x,l}^H) \right)^H \text{vec}(\hat{\mathbf{h}}_{j,z,l'} \hat{\mathbf{h}}_{j,x,l'}^H) \\ &\stackrel{(a)}{=} \sum_{l=1}^{\tilde{r}} \sum_{l'=1}^{\tilde{r}} (\hat{\mathbf{h}}_{i,x,l}^* \otimes \hat{\mathbf{h}}_{i,z,l})^H (\hat{\mathbf{h}}_{j,x,l'}^* \otimes \hat{\mathbf{h}}_{j,z,l'}) \\ &\stackrel{(b)}{=} \sum_{l=1}^{\tilde{r}} \sum_{l'=1}^{\tilde{r}} (\hat{\mathbf{h}}_{i,x,l}^T \hat{\mathbf{h}}_{j,x,l'}^*) (\hat{\mathbf{h}}_{i,z,l}^H \hat{\mathbf{h}}_{j,z,l'}). \end{aligned} \quad (23)$$

Here, “ $*$ ” denotes the conjugate operation, the equality (a) holds due to $\text{vec}(\mathbf{a}\mathbf{b}^H) = \mathbf{b}^* \otimes \mathbf{a}$, and the equality (b) comes from the fact $(\mathbf{a} \otimes \mathbf{b})^H (\mathbf{c} \otimes \mathbf{d}) = (\mathbf{a}^H \mathbf{c}) (\mathbf{b}^H \mathbf{d})$. Therefore, the Gram matrix $\hat{\mathbf{G}}$ can be further expressed as

$$\hat{\mathbf{G}} = \sum_{l=1}^{\tilde{r}} \sum_{l'=1}^{\tilde{r}} \left(\hat{\mathbf{H}}_{x,l}^H \hat{\mathbf{H}}_{x,l'} \right)^* \circ \left(\hat{\mathbf{H}}_{z,l}^H \hat{\mathbf{H}}_{z,l'} \right), \quad (24)$$

where $\hat{\mathbf{H}}_{x,l} = [\hat{\mathbf{h}}_{1,x,l}, \dots, \hat{\mathbf{h}}_{K,x,l}] \in \mathbb{C}^{M_x \times K}$, $\hat{\mathbf{H}}_{z,l} =$

$[\hat{\mathbf{h}}_{1,z,l}, \dots, \hat{\mathbf{h}}_{K,z,l}] \in \mathbb{C}^{M_z \times K}$, and “ $\circ$ ” denotes the Hadamard product. The computational complexity of (24) involves $\tilde{r}^2$ matrix multiplications, leading to an overall complexity of $\mathcal{O}(\tilde{r}^2 M_x K^2 + \tilde{r}^2 M_z K^2)$. ■

Moreover, we exploit the Hermitian structure of $\hat{\mathbf{G}}$ and the conjugate symmetry between path-pair contributions (l, l') and (l', l) . Specifically, the path-pair contribution terms satisfy

$$\mathbf{T}_{l',l} = \mathbf{T}_{l,l'}^H, \quad (25)$$

where $\mathbf{T}_{l,l'} = \left(\mathbf{H}_{x,l}^H \mathbf{H}_{x,l'} \right)^* \circ \left(\mathbf{H}_{z,l}^H \mathbf{H}_{z,l'} \right)$. Accordingly, it suffices to evaluate the path-pair terms for $l \leq l'$, where each $\mathbf{T}_{l,l'}$ is computed once and its conjugate transpose is reused to update $\hat{\mathbf{G}}$. This reduces the number of path-pair evaluations from $\tilde{r}^2$ to $\tilde{r}(\tilde{r} + 1)/2$, leading to a substantial complexity reduction without additional approximations. As for the final stage, the linear detection, the calculation of $\tilde{\mathbf{x}}_{\text{ZF}}$ or $\tilde{\mathbf{x}}_{\text{MMSE}}$ involves a $K \times K$ matrix inversion and two matrix-vector multiplications, resulting in a complexity of $\mathcal{O}(M_z M_x K + K^3)$.

However, spherical wave propagation gives rise to multiple spatial directions across the antenna array, which increases the required approximation rank $\tilde{r}$ of $\mathbf{H}_k$ and significantly raises the computational complexity of the global LMaFit.

IV. BLOCK-WISE LMAFIT SCHEME

In this section, we introduce the block-wise LMaFit to achieve further complexity reduction. Then we analyze the computational complexity and convergence behaviors to demonstrate the superiority of block-wise LMaFit compared to the global LMaFit.

A. Algorithm Description

To achieve further complexity reduction, we develop a block-wise LMaFit framework. In particular, the large antenna array is partitioned into B subarrays according to the estimated VR of each user, such that the subchannel matrix $\mathbf{H}_k$ can be decomposed as

$$\mathbf{H}_k = \sum_{b=1}^B \mathbf{M}_k^{(b)} \mathbf{H}_k^{(b)} \mathbf{N}_k^{(b)H}. \quad (26)$$

Here, $\mathbf{H}_k^{(b)} \in \mathbb{C}^{M_{z_b} \times M_{x_b}}$ denotes the local subchannel matrix associated with the b -th subchannel, and $\mathbf{M}_k^{(b)} \in \mathbb{C}^{M_z \times M_{z_b}}$ and $\mathbf{N}_k^{(b)} \in \mathbb{C}^{M_x \times M_{x_b}}$ are row and column selection matrices, which embed $\mathbf{H}_k^{(b)}$ into the full antenna domain.

Based on this block-wise formulation, the block-wise LMaFit problem for the b -th subchannel to obtain a rank- $\tilde{r}_b$ approximation $\hat{\mathbf{H}}_k^{(b)}$ of $\mathbf{H}_k^{(b)}$ is given by

$$\min_{\hat{\mathbf{H}}_{k,z}^{(b)}, \hat{\mathbf{H}}_{k,x}^{(b)}} F\left(\hat{\mathbf{H}}_{k,z}^{(b)}, \hat{\mathbf{H}}_{k,x}^{(b)}\right), \quad (27)$$

where

$$F\left(\hat{\mathbf{H}}_{k,z}^{(b)}, \hat{\mathbf{H}}_{k,x}^{(b)}\right) = \frac{1}{2} \left\| \mathbf{H}_k^{(b)} - \hat{\mathbf{H}}_{k,z}^{(b)} \hat{\mathbf{H}}_{k,x}^{(b)H} \right\|_F^2 \quad (28)$$

with $\hat{\mathbf{H}}_{k,z}^{(b)} \in \mathbb{C}^{M_{z_b} \times \tilde{r}_b}$ and $\hat{\mathbf{H}}_{k,x}^{(b)} \in \mathbb{C}^{M_{x_b} \times \tilde{r}_b}$. To sum up, the proposed block-wise LMaFit scheme is outlined in Algorithm 2.

Algorithm 2 Block-wise LMaFit Algorithm

Input: $B, \{\mathbf{H}_k^{(b)}, \mathbf{M}_k^{(b)}, \mathbf{N}_k^{(b)}\}_{b=1}^B, T_{\max}$
Output: Approximated matrix $\mathbf{H}_k \approx \sum_{b=1}^B \mathbf{M}_k^{(b)} \hat{\mathbf{H}}_k^{(b)} \mathbf{N}_k^{(b)}$
1: **for** $b = 1$ to B **do**
2: **Call** Algorithm 1($\mathbf{H}_k^{(b)}, T_{\max}$) to obtain $\hat{\mathbf{H}}_k^{(b)}$
3: **end for**

Moreover, each block channel typically admits a smaller effective rank, owing to the combined effects of VR-based block partitioning and the spatial stationarity within each subarray [7]. Consequently, block-wise LMaFit can operate with a smaller approximation rank $\tilde{r}_b$ than the global scheme, thereby enabling a further reduction in the rank-dependent computational complexity of Gram matrix construction.

B. Convergence Analysis

To establish the convergence consistency between the proposed block-wise LMaFit and the global LMaFit, we examine their stationary-point behavior in the following Corollary.

Corollary 1. *Given the decomposition in (26), the block-wise LMaFit converges to a stationary point of the same global objective as the global LMaFit.*

Proof: Based on Lemma 1, for every block b , the factor matrices $(\hat{\mathbf{H}}_{k,z}^{(b,t)}, \hat{\mathbf{H}}_{k,x}^{(b,t)})$ generated by block-wise LMaFit converge to the stationary point of the subproblem in (28), i.e.,

$$\lim_{t \rightarrow \infty} \nabla_{\hat{\mathbf{H}}_{k,z}^{(b,t)}, \hat{\mathbf{H}}_{k,x}^{(b,t)}} F\left(\hat{\mathbf{H}}_{k,z}^{(b,t)}, \hat{\mathbf{H}}_{k,x}^{(b,t)}\right) = \mathbf{0}, \quad \forall b. \quad (29)$$

By embedding the block-wise factors into the global antenna domain, the global factor matrices are constructed as

$$\hat{\mathbf{H}}_{k,z}^{(t)} = \sum_{b=1}^B \mathbf{M}_k^{(b,t)} \hat{\mathbf{H}}_{k,z}^{(b,t)}, \quad \hat{\mathbf{H}}_{k,x}^{(t)} = \sum_{b=1}^B \mathbf{N}_k^{(b,t)} \hat{\mathbf{H}}_{k,x}^{(b,t)}. \quad (30)$$

Since the global LMaFit objective is additively separable with respect to the block partition, the block-wise stationary conditions jointly implies

$$\lim_{t \rightarrow \infty} \nabla_{\hat{\mathbf{H}}_{k,z}^{(t)}, \hat{\mathbf{H}}_{k,x}^{(t)}} F\left(\hat{\mathbf{H}}_{k,z}^{(t)}, \hat{\mathbf{H}}_{k,x}^{(t)}\right) = \mathbf{0}, \quad (31)$$

which coincides with (22). This completes the proof. ■

Corollary 1 implies that both schemes converge to stationary points of the same global objective, while the corresponding stationary points may differ. To further compare the per-iteration residual reduction behaviors of the block-wise and global LMaFit schemes, we analyze the ideal case where, at each iteration t , the low-rank factors achieve the optimal approximation of the current matrix $\hat{\mathbf{H}}_k^{(t)}$ in the Frobenius norm sense, i.e., the approximation error attains the minimum achievable by TSVD under the given rank. This leads to the following Theorem.

Theorem 1. *For the near-field channel matrix in (2) and a fixed residual matrix $\mathbf{S}^{(t)}$, under optimal matrix approximations, the*

TABLE I
COMPLEXITY COMPARISON OF DETECTION ALGORITHMS FOR $K = 128$ AND DIFFERENT $M_z = M_x$

Algorithm	Computational Complexity	$M_z = 32$ (FLOPs)	$M_z = 48$ (FLOPs)
Conventional ZF	$2M_z M_x K^2 + 2M_z M_x K + K^3$	39671100	82269500
Global LMaFit-ZF	$\sum_{t=1}^{T_{\max}} K \left(2\tilde{r}^{(t)} M_z M_x + \tilde{r}^{(t)2} M_z + 4M_z M_x + M_z \tilde{r}^{(t)} \right) + \tilde{r}^{(T_{\max})2} (M_z + M_x + 1) K^2 + 2M_z M_x K + K^3$	15871400	25975100
Block-wise LMaFit-ZF	$\sum_{t=1}^{T_{\max}} \sum_{b=1}^B K \left(2\tilde{r}_b^{(t)} M_{z_b} M_{x_b} + \tilde{r}_b^{(t)2} M_{z_b} + 4M_{z_b} M_{x_b} + M_{z_b} \tilde{r}_b^{(t)} \right) + \sum_{b=1}^B \left(\tilde{r}_b^{(T_{\max})2} (M_{z_b} + M_{x_b} + 1) K^2 + 2M_{z_b} M_{x_b} K \right) + K^3$	6480080	7406640

residual $\|\mathbf{S}^{(t+1)}\|_F^2$ achieved by block-wise LMaFit is no larger than that of global LMaFit, namely,

$$\|\mathbf{S}_{\text{block}}^{(t+1)}\|_F^2 \leq \|\mathbf{S}_{\text{global}}^{(t+1)}\|_F^2. \quad (32)$$

Proof: To begin with, the residual norm produced by global LMaFit at iteration $t + 1$ can be expressed as [3]

$$\|\mathbf{S}_{\text{global}}^{(t+1)}\|_F^2 = \|(\mathbf{I} - \mathbf{Q}^{(t+1)})\mathbf{S}^{(t)}(\mathbf{I} - \mathbf{P}^{(t)})\|_F^2, \quad (33)$$

where $\mathbf{Q}^{(t+1)}$ denotes the column projection matrix of rank- $\tilde{r}^{(t+1)}$ constructed at iteration $t + 1$, and $\mathbf{P}^{(t)}$ is the row projection matrix obtained at iteration t . Moreover, the residual norm of block-wise LMaFit at iteration $t + 1$ is given by

$$\begin{aligned} \|\mathbf{S}_{\text{block}}^{(t+1)}\|_F^2 &= \sum_{b=1}^B \|\mathbf{S}^{(b,t+1)}\|_F^2 \\ &= \sum_{b=1}^B \left\| (\mathbf{I} - \mathbf{Q}^{(b,t+1)})\mathbf{S}^{(b,t)}(\mathbf{I} - \mathbf{P}^{(b,t)}) \right\|_F^2, \end{aligned} \quad (34)$$

where $\mathbf{Q}^{(b,t+1)}$ is the rank- $\tilde{r}_b^{(t+1)}$ column projection matrix for block b at iteration $t + 1$, and $\mathbf{P}_t^{(b,t)}$ denotes the row projection matrix for block b at iteration t .

Under the optimal matrix approximation assumption, by the Eckart–Young Theorem [8], we can obtained

$$\begin{aligned} \left\| (\mathbf{I} - \mathbf{Q}^{(t+1)})\mathbf{S}^{(t)}(\mathbf{I} - \mathbf{P}^{(t)}) \right\|_F^2 &= \sum_{i > \tilde{r}^{(t+1)}} \sigma_i^2(\mathbf{S}^{(t)}), \\ \sum_{b=1}^B \left\| (\mathbf{I} - \mathbf{Q}^{(b,t+1)})\mathbf{S}^{(b,t)}(\mathbf{I} - \mathbf{P}^{(b,t)}) \right\|_F^2 &= \sum_{b=1}^B \sum_{i > \tilde{r}_b^{(t+1)}} \sigma_i^2(\mathbf{S}^{(b,t)}), \end{aligned} \quad (35)$$

where $\sigma_i^2(\cdot)$ denotes the singular values in descending order.

Since the blocks are non-overlapping and all entries outside these blocks are zero, the matrix $\mathbf{S}^{(t)}$ can be permuted into a block-diagonal form by unitary row and column permutation matrices $\mathbf{V}_r$ and $\mathbf{V}_c$, i.e.,

$$\tilde{\mathbf{S}}^{(t)} = \mathbf{V}_r \mathbf{S}^{(t)} \mathbf{V}_c^H = \text{blkdiag}(\mathbf{S}^{(1,t)}, \dots, \mathbf{S}^{(B,t)}, \mathbf{0}). \quad (36)$$

The singular values are invariant under such transformation, thus, the global singular values of $\mathbf{S}^{(t)}$ are obtained by the sorted union of those of $\{\mathbf{S}^{(b,t)}\}_{b=1}^B$. Moreover, under the proposed rank adaptation strategy, the rank condition $\tilde{r}^{(t+1)} \leq \sum_{b=1}^B \tilde{r}_b^{(t+1)}$ is satisfied at each iteration by the rank update rule. It follows that

$$\sum_{i > \tilde{r}^{(t+1)}} \sigma_i^2(\mathbf{S}^{(t)}) \geq \sum_{b=1}^B \sum_{i > \tilde{r}_b^{(t+1)}} \sigma_i^2(\mathbf{S}^{(b,t)}), \quad (37)$$

and therefore,

$$\|\mathbf{S}_{\text{block}}^{(t+1)}\|_F^2 \leq \|\mathbf{S}_{\text{global}}^{(t+1)}\|_F^2. \quad (38)$$

The block-wise LMaFit may achieve improved practical performance when the channel exhibits spatially varying low-rank structures across VR blocks, where a global model may not fully capture local variations. The theoretical results will be validated in Section V through numerical simulations. ■

C. Complexity Analysis

We now evaluate the computational complexity of the proposed block-wise LMaFit. For simplicity, the same approximation rank $\tilde{r}_b$ is assumed for all K users in the complexity analysis.

Since each block-wise subproblem shares the same LMaFit structure as the global formulation, the per-iteration complexity follows directly from Section III-B, with the matrix dimensions replaced accordingly. Specifically, during the LMaFit processing stage, the updates of $\hat{\mathbf{H}}_{k,z}^{(b)}$ and $\hat{\mathbf{H}}_{k,x}^{(b)}$ demand a complexity of $\mathcal{O}(M_{z_b} M_{x_b} + M_{z_b} \tilde{r}_b)$. Then, the residual-based rank adaptation procedure incurs a computational complexity of $\mathcal{O}(M_{z_b} M_{x_b})$. In the second stage, based on Proposition 1, the complexity of computing the Gram matrix can be substantially reduced as follows. Here, the proof is omitted where the related details can be found in Proposition 1.

Propositon 2. *Given the rank- $\tilde{r}_b$ factorized approximation obtained by Algorithm 2, the computational complexity of the Gram matrix $\hat{\mathbf{G}} = \hat{\mathbf{H}}^H \hat{\mathbf{H}}$ can be reduced from $\mathcal{O}(M_z M_x K^2)$ to $\mathcal{O}(\sum_{b=1}^B (\tilde{r}_b^2 M_{z_b} K^2 + \tilde{r}_b^2 M_{x_b} K^2))$.*

As for the subsequent linear detection stage, the detection procedure is identical to the global LMaFit-based scheme, and its computational complexity remains $\mathcal{O}(M_z M_x K + K^3)$.

To summarize, the overall computational complexity of the proposed LMaFit-ZF is compared in Table I, where $\tilde{r}, \tilde{r}_b \leq T_{\max}$ under the adopted rank adaptation strategy. It can be observed that the complexity advantage of the proposed detection becomes more pronounced for large-scale UPA configurations, particularly when the channel matrices exhibit a low effective rank.

V. SIMULATION

In this section, we evaluate the performance of the proposed LMaFit-based linear detection for uplink near-field XL-MIMO systems. The XL-MIMO system employs a 32×32 UPA, divided into 8×4 sub-arrays, where each user is visible to $B = 4$ subarrays, and the number of paths is $L = 8$. The users' azimuth

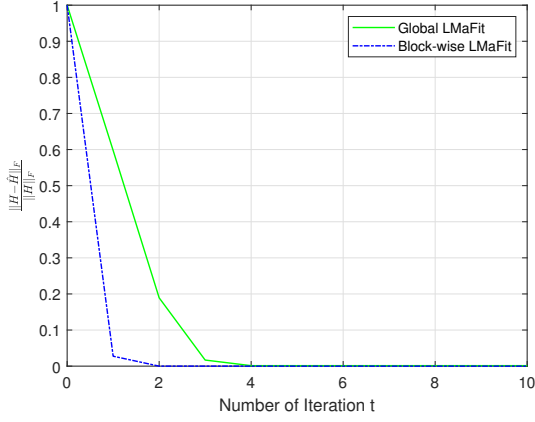


Fig. 1. Comparison of the convergence performance of two LMaFit methods

AoAs are uniformly distributed in $[-\pi, \pi]$ and the elevation AoAs are uniformly distributed in $[\pi/6 - \delta, \pi/6 + \delta]$.

Fig. 1 compares the convergence behaviors of the global and block-wise LMaFit in 1024×32 XL-MIMO systems with $\delta = 0^\circ$. Both methods employ the rank adaptation (RA) strategy described in Section III-B. Notably, although the global LMaFit and block-wise LMaFit eventually achieve comparable approximation accuracy, the block-wise LMaFit exhibits a significantly faster convergence rate, which is consistent with Theorem 1.

Fig. 2 evaluates the performance of the proposed LMaFit-based ZF detection scheme in 1024×32 XL-MIMO systems with $\delta = \pi/6$. For a fair comparison, we consider two rank-increasing strategies: the RA strategy and a heuristic residual-stagnation-based (RS) strategy proposed in [3]. Clearly, block-wise LMaFit-based ZF with $t = 1$ already achieves performance close to that of the ZF, and significantly outperforms global LMaFit-based ZF. In addition, it also exhibits better performance compared to the user partition-based ZF algorithm proposed in [9].

Fig. 3 further compares the computational complexity under the same system settings as in Fig. 2. By exploiting the near-field VR characteristics through block-wise processing, the proposed scheme achieves a markedly lower complexity than both the ZF and the global LMaFit-based ZF.

VI. CONCLUSION

In this paper, we developed a global LMaFit scheme with a rank adaptation strategy to enable low-complexity Gram matrix construction for linear detection. Building upon this, a block-wise LMaFit scheme was further proposed to substantially reduce the computational complexity by exploiting localized low-rank structures within subarrays. Moreover, we showed that the block-wise LMaFit attains stationary solutions of the same global objectives as the global LMaFit, while achieving faster convergence.

VII. ACKNOWLEDGMENT

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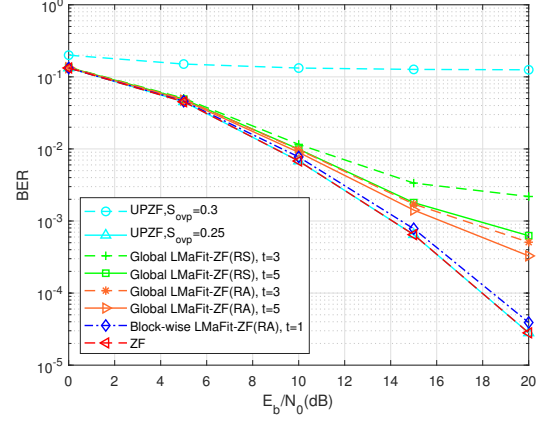


Fig. 2. BER versus E_b/N_0 for the uncoded XL-MIMO system with 16-QAM.

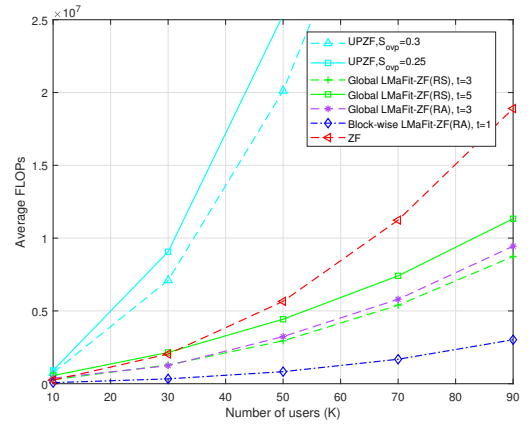


Fig. 3. Complexity versus the number of users K .

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